

Overview of common therapeutic options

Table 4 provides a summary of evidence level, efficacy in preventing progression and/or improving androgenetic alopecia (AGA), safety, and practicability for the most commonly used therapeutic interventions. The table is intended to offer a preliminary orientation. However, an individualized treatment approach is essential, taking into account each patient's clinical condition, expectations, needs, and concerns. The optimal therapy should be decided jointly by the physician and patient, considering the best available evidence, expected outcomes, ease of use, and likelihood of adherence.

Minoxidil

Efficacy – males

Forty-eight studies evaluating minoxidil's efficacy in men with AGA met the inclusion criteria. The majority of these studies provided grade A2 ($n = 22$) or grade B ($n = 18$) evidence, with eight rated as grade C, resulting in an overall evidence level 1.

- Minoxidil 2% solution applied twice daily is effective in preventing progression and improving AGA in the frontotemporal and vertex regions (evidence level 1).
- Minoxidil 5% solution or foam applied twice daily is more effective than the 2% formulation (evidence level 2).

One study using oral minoxidil at 5 mg/day in men showed increased hair count, though the improvement was only marginally greater than that seen with topical minoxidil. However, oral administration was associated with

notable side effects, including hypertrichosis (93%), pedal edema (10%), and ECG changes (10%).

Efficacy – females

Nineteen studies on topical minoxidil in women with AGA were included. Seven provided grade A2 evidence, nine grade B, and three grade C, leading to an overall **evidence level 1**.

- Minoxidil 2% solution applied twice daily is effective in preventing progression and improving AGA in women (evidence level 1).
- Minoxidil 5% solution or foam applied once daily shows comparable efficacy to the 2% solution applied twice daily (evidence level 2).

Instructions for use, practicability, and side effects

- Apply 1 mL of solution (using a pipette) or half a cap of foam to dry hair and scalp, twice daily in men and once daily in women.
- When using a spray, ensure even distribution over affected areas.
- Treatment response should be evaluated after 6 months. If effective, ongoing use is required to maintain benefits.

Patients should be informed about transient increased telogen shedding during the first few months of treatment. Discontinuation of minoxidil typically leads to resumption of hair loss.

Common side effects: - **Hypertrichosis**, likely due to local spread, excessive application, or individual sensitivity. - **Irritant or allergic contact dermatitis**, more frequently associated with the 5% solution due to its propylene glycol content. Allergic reactions to propylene glycol or minoxidil can be confirmed via patch testing.

Due to insufficient safety data, topical minoxidil should be discontinued during pregnancy and lactation.

Combination therapies – males and females

Nine studies in men and six in women compared topical minoxidil monotherapy with combination treatments.

- In men: Three studies provided grade A2 evidence and six grade B. However, due to limited data per specific combination (only one or two studies each), the overall evidence level is **3**.
- In women: Three studies provided grade B evidence and three grade C, also resulting in **evidence level 3**.

For detailed results on specific combinations, refer to the full guideline document.